Assignment 01: Git, Gitea, GitHub, Git LFS, and GitHub Pages

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Operating System: Ubuntu Server (VMware)

1. Introduction

The objective of this assignment was to gain practical experience with Git, Gitea, GitHub, Git Large File Storage (LFS), and GitHub Pages. The assignment involved setting up a self-hosted Git server, managing repositories, uploading large files, and deploying a personal portfolio website.

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2. Task 1: Gitea Installation and Repository Setup

Procedure

1. Installed Docker and Docker Compose on Ubuntu Server.
2. Downloaded the Gitea configuration files.
3. Started Gitea using Docker Compose.
4. Accessed the Gitea web interface and created an account.
5. Created a public repository named Assignment01.
6. Initialized a local Git repository and created README.md containing student information.
7. Committed the README and pushed it to Gitea.

Commands used:

```
docker compose up -d

docker compose ps
```

mkdir ~/Assignment01

cd ~/Assignment01

git init

git branch -M main

git add README.md

git commit -m "Add student information"

git push -u gitea main

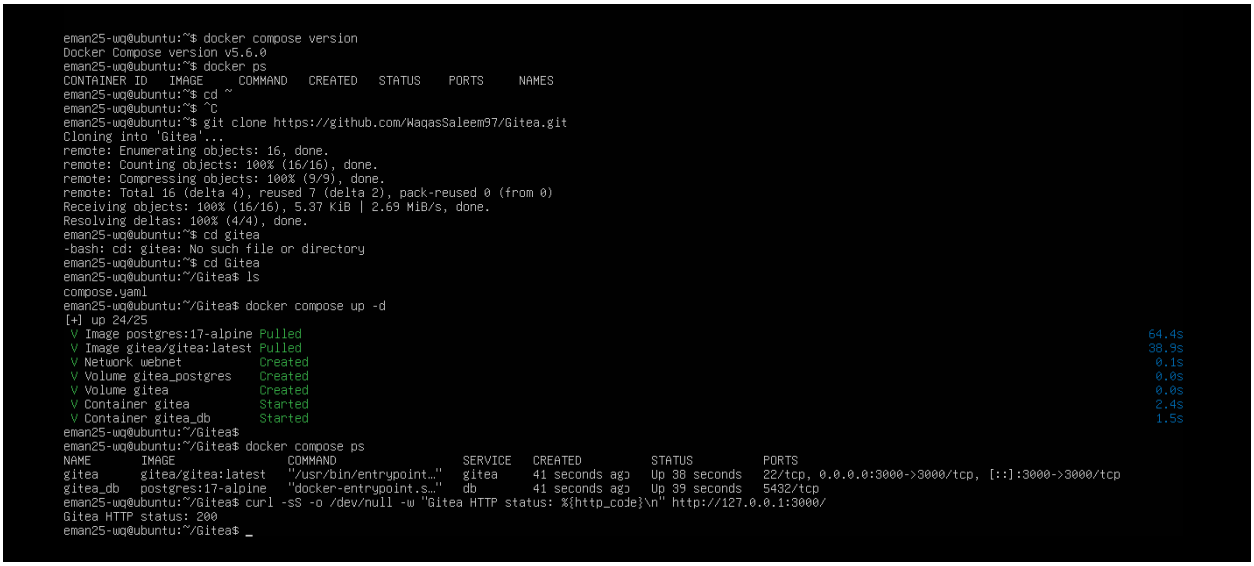


Figure 1: Gitea running successfully using Docker.

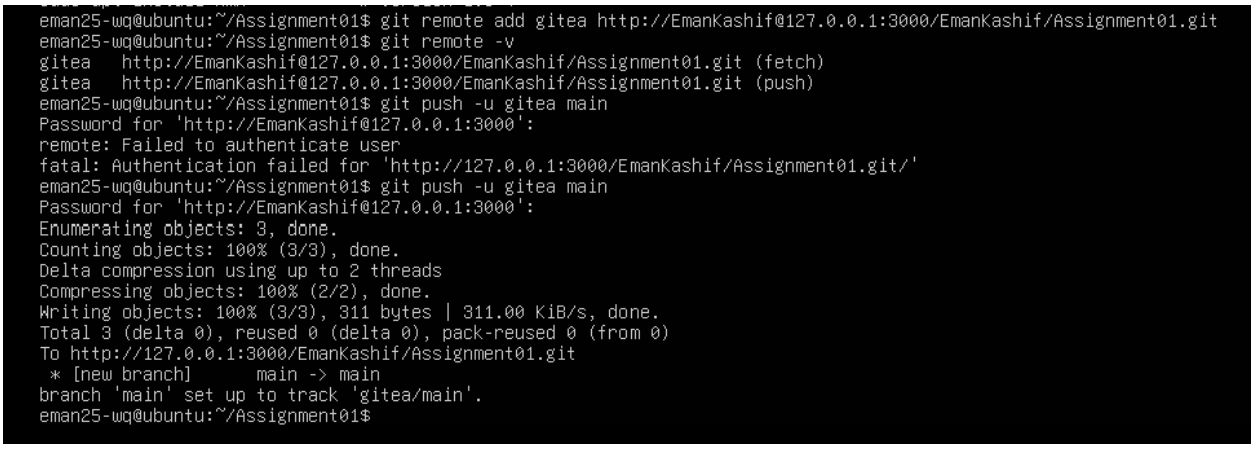


Figure 2: Successful push from Ubuntu to Gitea.

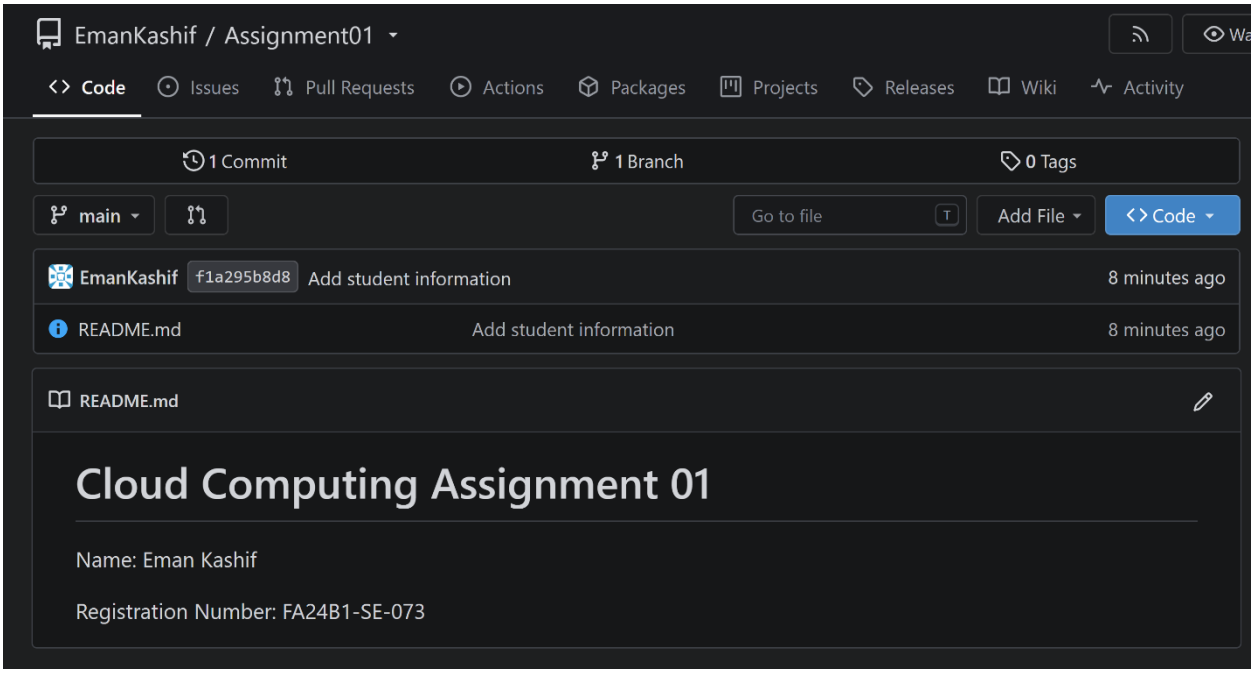


Figure 3: Public Gitea repository containing README.md.

Result: Gitea was successfully deployed, and the local Git repository was uploaded to the self-hosted server.

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3. Task 2: GitHub Repository Integration

Procedure

1. Created a public GitHub repository named Assignment01.
2. Connected the existing Ubuntu repository to GitHub using a second remote.
3. Authenticated using GitHub CLI.
4. Pushed the local repository to GitHub.
5. Verified that the README was displayed correctly.

Commands used:

git remote add github https://github.com/eman25-wq/Assignment01.git

git remote -v

git push -u github main

```
eman25-wq@ubuntu:~/Gitea$ cd ~/Assignment01
eman25-wq@ubuntu:~/Assignment01$ git remote -v
gitea   http://EmanKashif@127.0.0.1:3000/EmanKashif/Assignment01.git (fetch)
gitea   http://EmanKashif@127.0.0.1:3000/EmanKashif/Assignment01.git (push)
eman25-wq@ubuntu:~/Assignment01$ git remote add github https://github.com/eman25-uq/Assignment01.git
eman25-wq@ubuntu:~/Assignment01$ git remote -v
gitea   http://EmanKashif@127.0.0.1:3000/EmanKashif/Assignment01.git (fetch)
gitea   http://EmanKashif@127.0.0.1:3000/EmanKashif/Assignment01.git (push)
github  https://github.com/eman25-uq/Assignment01.git (fetch)
github  https://github.com/eman25-uq/Assignment01.git (push)
eman25-wq@ubuntu:~/Assignment01$ _
```

Figure 4: Git remote configuration showing Gitea and GitHub.

```
eman25-wq@ubuntu:~/Assignment01$ git remote set-url github https://github.com/eman25-wq/Assignment01.git
eman25-wq@ubuntu:~/Assignment01$ git push -u github main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 311 bytes | 311.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/eman25-wq/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
eman25-wq@ubuntu:~/Assignment01$
```

Figure 5: Successful push to GitHub.

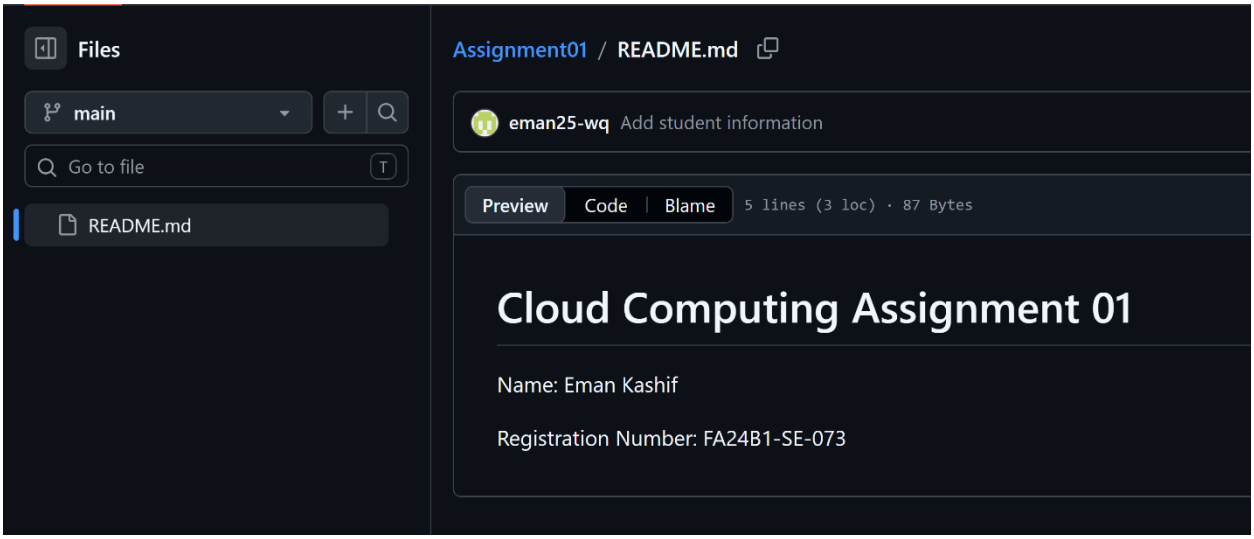


Figure 6: GitHub repository displaying student information.

Result: The same local Git repository was successfully connected to both Gitea and GitHub.

Repository: Assignment01 on GitHub

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4. Task 3: Git Large File Storage (LFS)

Procedure

1. Installed and initialized Git LFS.
2. Configured Git LFS to track .bin files.
3. Created three binary files larger than 100 MiB.
4. Added and committed the files.
5. Uploaded the files to GitHub using Git LFS.

Commands used:

git lfs install

git lfs track "*.bin"

truncate -s 101M large-file-1.bin

truncate -s 102M large-file-2.bin

truncate -s 103M large-file-3.bin

git add .gitattributes *.bin

git lfs ls-files --size

git commit -m "Add three large files using Git LFS"

git push github main

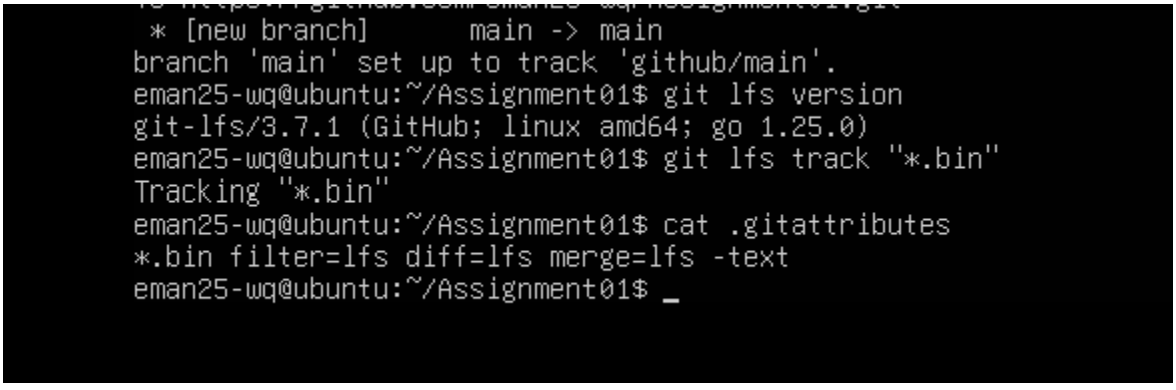


Figure 7: Git LFS installation and configuration.

```
eman25-wq@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
eman25-wq@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
eman25-wq@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
eman25-wq@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
eman25-wq@ubuntu:~/Assignment01$ git add .gitattributes *.bin
eman25-wq@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
eman25-wq@ubuntu:~/Assignment01$ _
```

Figure 8: Three large binary files tracked using Git LFS.

```
Try: sudo apt install <deb name>
eman25-wq@ubuntu:~/Assignment01$ git push github main
Uploading LFS objects:   0% (0/3), 20 MB | 686 KB/s
nUploading LFS objects:   0% (0/3), 57 MB | 872 KB/s
eUploading LFS objects:   0% (0/3), 58 MB | 872 KB/s
xUploading LFS objects:   0% (0/3), 58 MB | 872 KB/s
Uploading LFS objects: 100% (3/3), 463 MB | 1.6 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 800 bytes | 800.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/eman25-wq/Assignment01.git
   f1a295b..fd7c85a  main -> main
eman25-wq@ubuntu:~/Assignment01$
eman25-wq@ubuntu:~/Assignment01$ _
```

Figure 9: Successful upload of large files to GitHub.

Result: Three files of 101 MiB, 102 MiB, and 103 MiB were created and uploaded using Git LFS.

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5. Task 4: Personal Portfolio Using GitHub Pages

Procedure

1. Created a public repository named eman25-wq.github.io.
2. Created index.html for the website structure.
3. Created styles.css for styling.
4. Added the following portfolio sections:

- About Me
- Education
- Skills
- Projects

5. Configured GitHub Pages to deploy from the main branch and root directory.
6. Opened the published website to verify deployment.

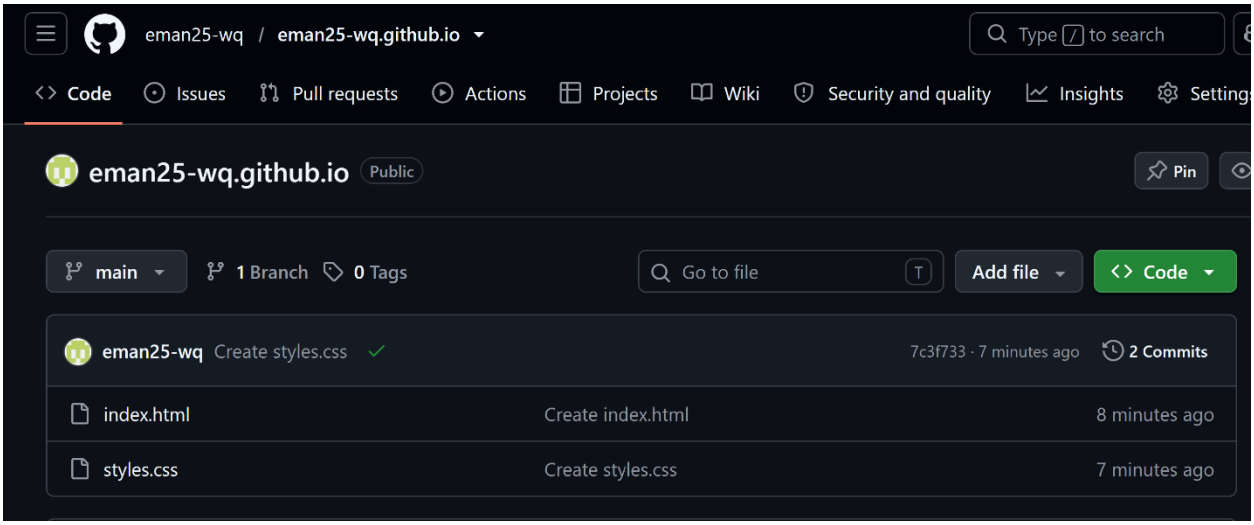


Figure 10: Portfolio repository containing HTML and CSS files.

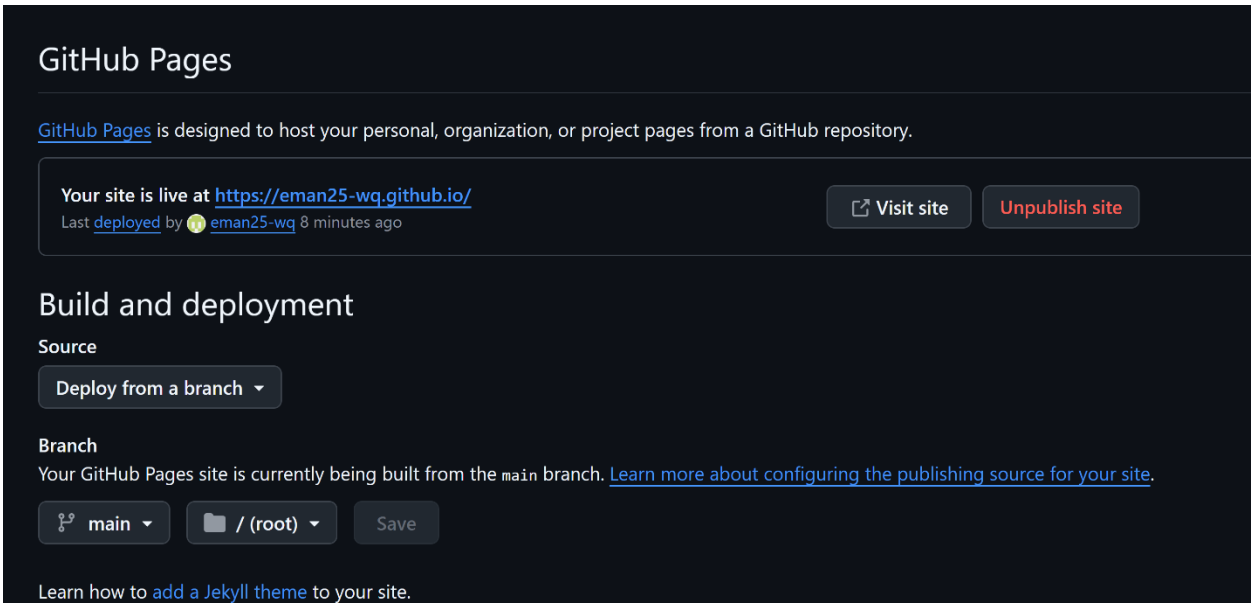


Figure 11: GitHub Pages deployment configuration.

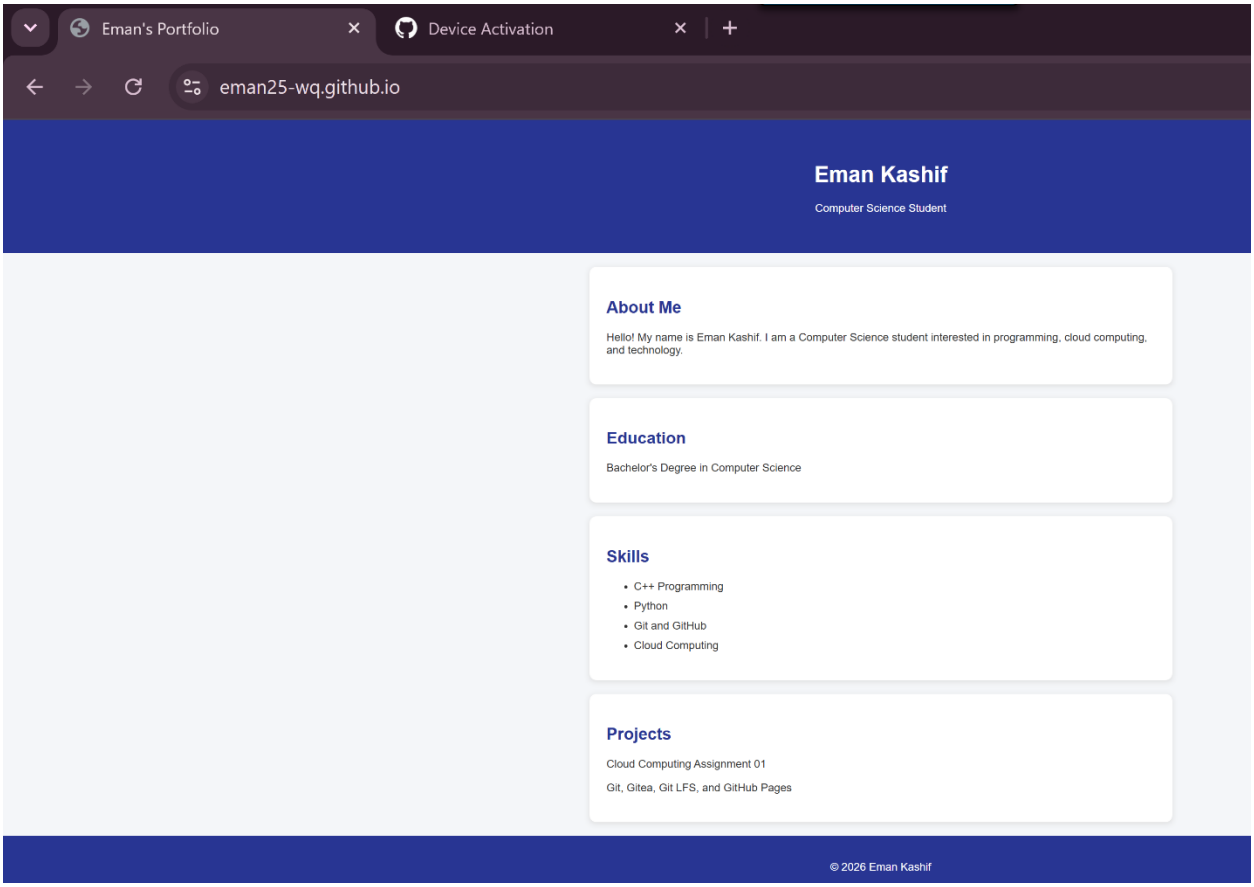


Figure 12: Successfully deployed personal portfolio website.

Result: The personal portfolio website was successfully published using GitHub Pages.

Live Website: Eman's Portfolio

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6. Conclusion

This assignment provided practical experience in version control, repository management, and website deployment. Gitea was used for self-hosted repository management, GitHub for remote repository hosting, Git LFS for handling large files, and GitHub Pages for publishing a personal portfolio. All four technical tasks were completed successfully.